

Experiment # 07

Introduction to Routers and Basic Commands

Objective:

The objective of this lab to learn the introduction to router and learn basic router commands.

Software Tool:

Cisco Packet Tracer

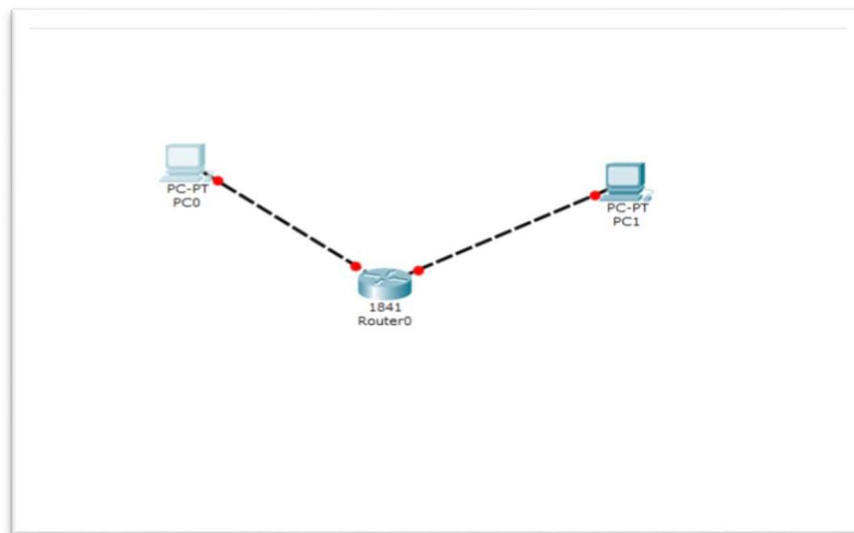
Theory:

A router is a networking device that connects different networks and forwards data packets between them. Routers operate at the **Network Layer** (Layer 3) of the OSI model and are essential for routing traffic between different IP subnets, which can be either on the same or different physical networks. Routers have multiple interfaces, each connected to a different network or subnet, and they use routing tables to determine the best path for data forwarding.

Procedure:

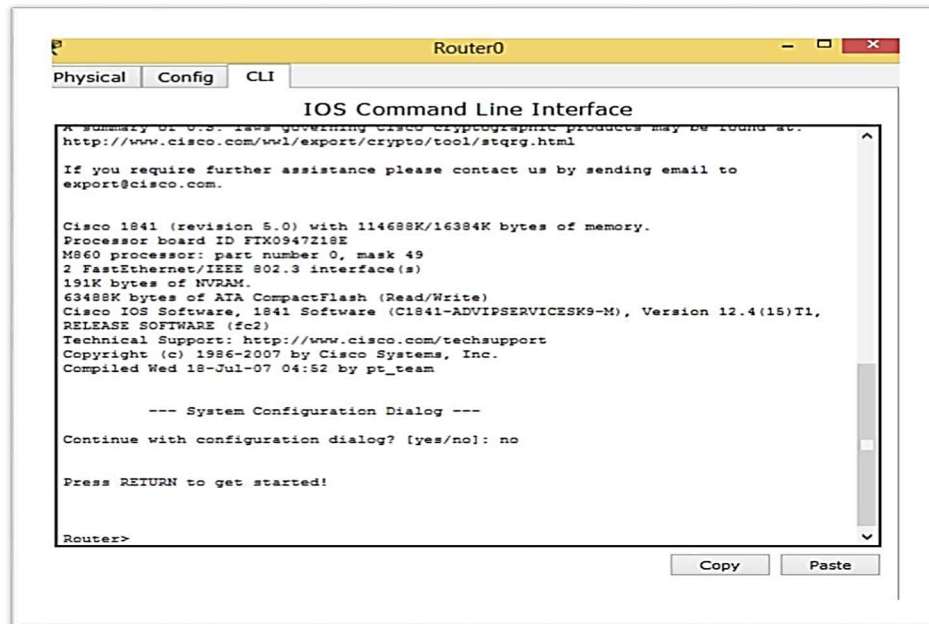
Physical networks:

In physical network we connect two pc with a router.



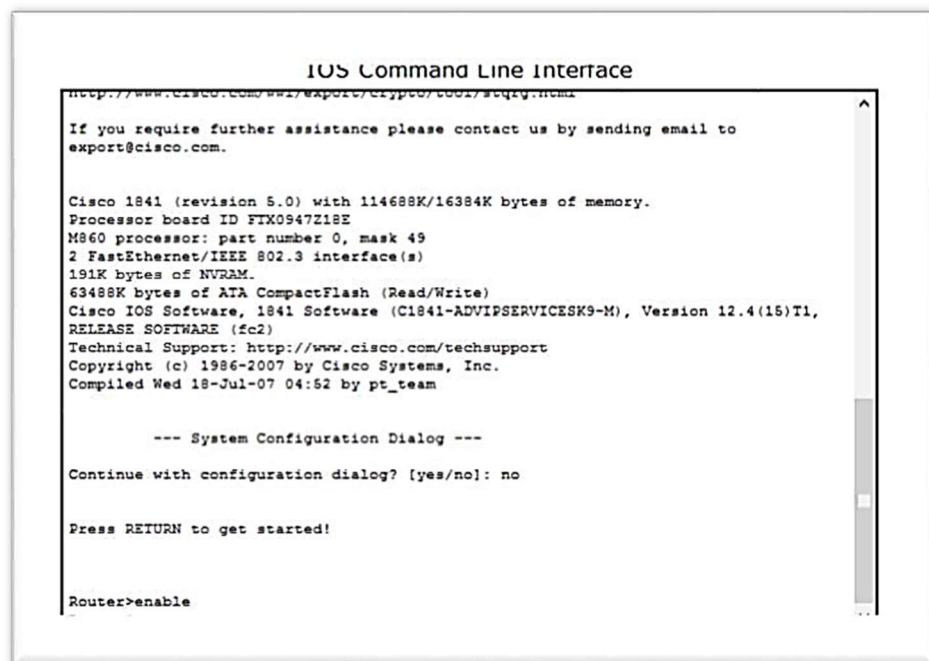
User exec mode:

We use IOS command line interface. When we open this command line interface we are in user exec mode. The user EXEC commands are a subset of the privileged EXEC commands.



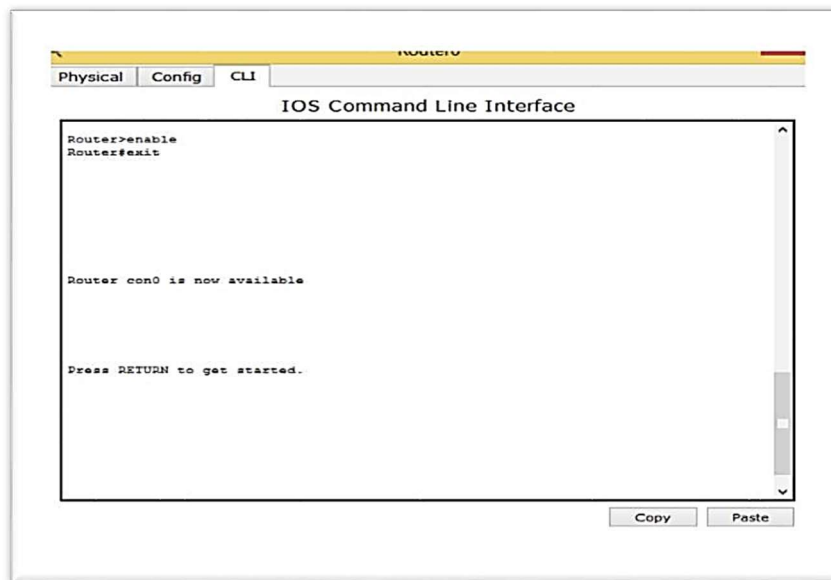
Privileged mode:

We are in user mode. But by using the command of "ENABLE" we move in privileged mode.



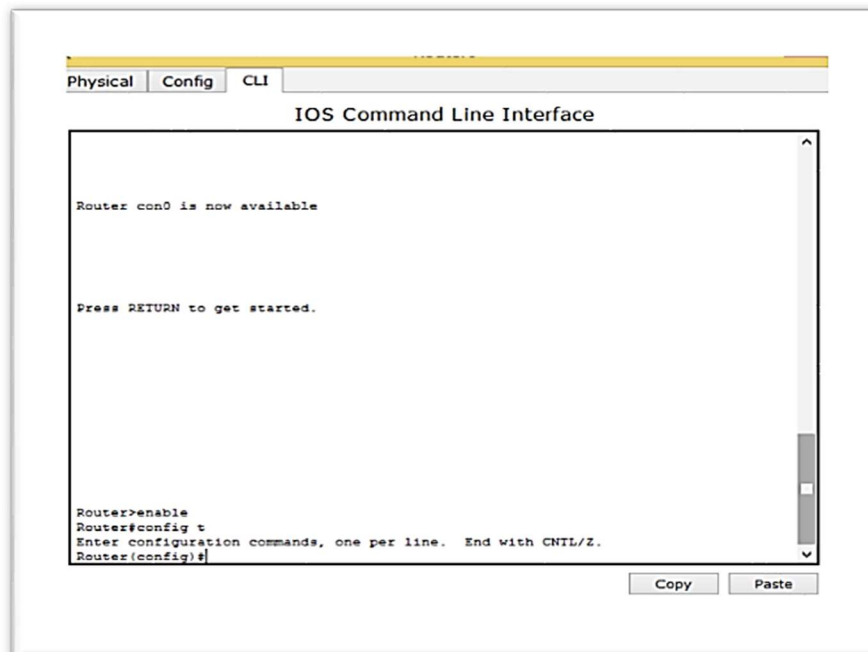
Move from privileged to user mode:

By using the command of “exit” we can again move from privileged to user exe mode.



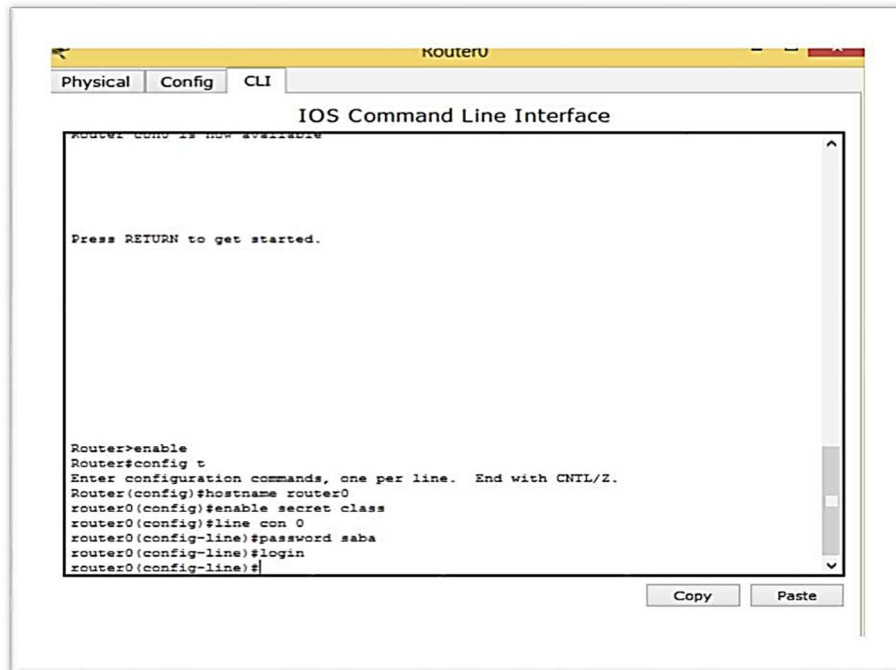
Move from privileged mode to global configuration mode:

By using the command of “config t” we move from privileged mode to global configuration mode.



To set password and login router in global mode:

We set the password in global mode by using password command.



Line Console password is to protect user mode. To protect privilege mode do the following steps:

1. Secret Password:

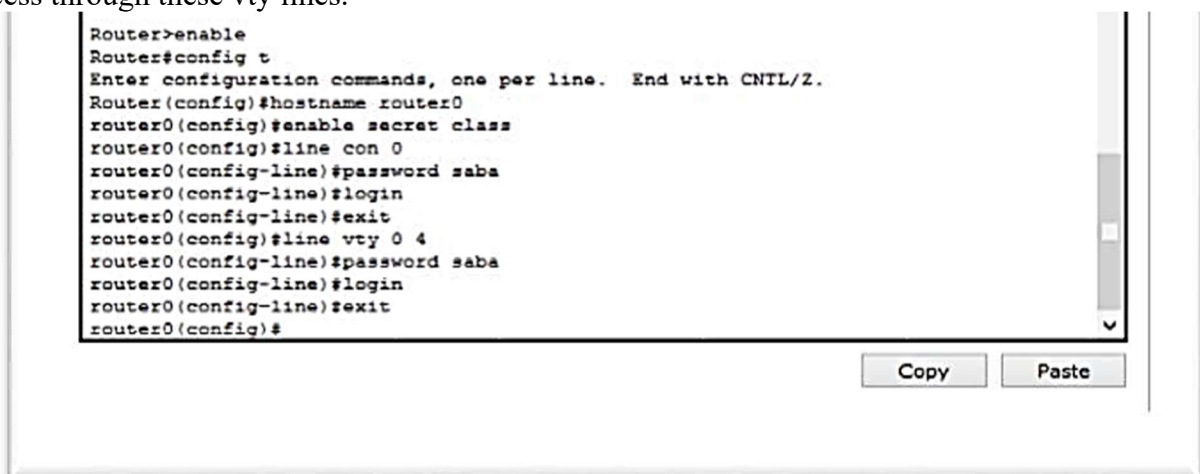
Router(config)#**enable secret** YOUR_PASSWORD

2. Enable Password (Less Secure):

Router(config)#**enable password** YOUR_PASSWORD

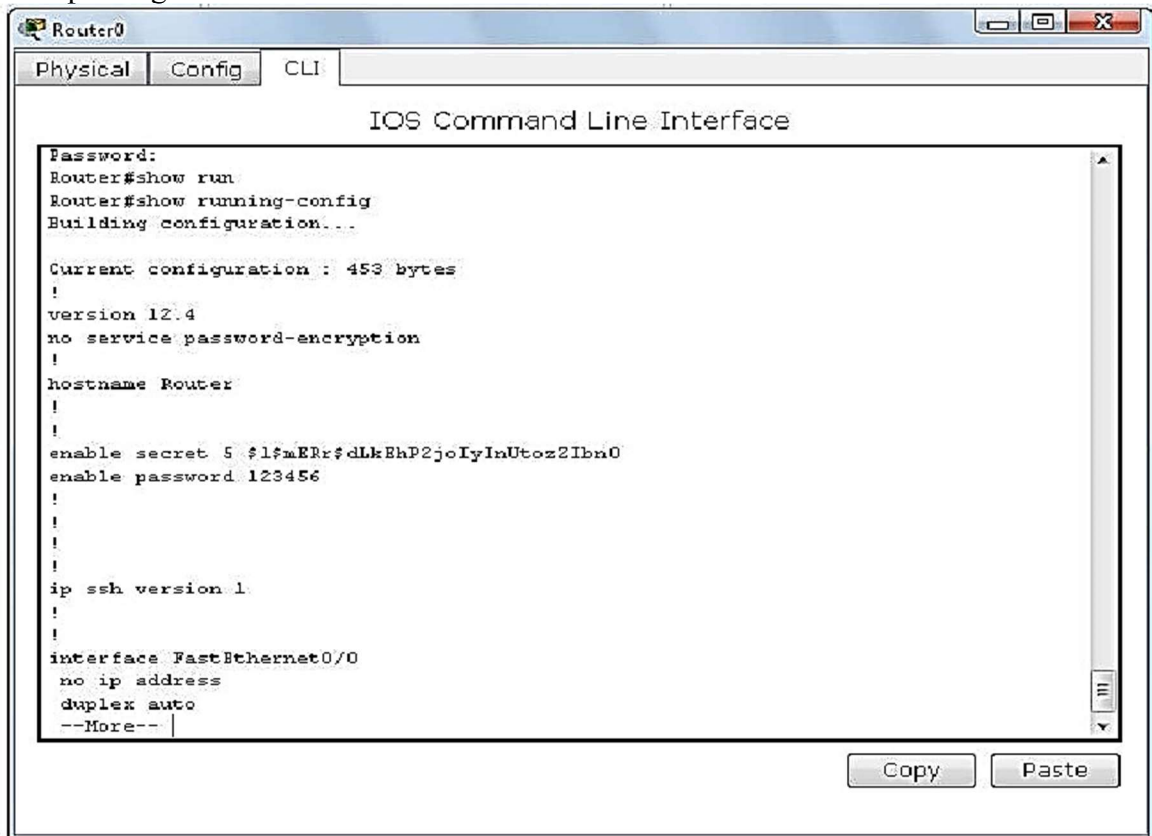
Configuration mode:

In global configuration mode five users simultaneously use telnet to remotely access host by using any of five virtual terminal lines. Now when someone tries to connect to the device remotely through Telnet using vty lines 0 to 4, they will be prompted to enter the password "saba" for authentication. The **login** command ensures that password-based authentication is required for access through these vty lines.



Show All the Settings:

Before saving all configurations, see all the settings you have recently made so incase of wrong setting you can re-correct it and the save it. To see all the settings, type `show running-config` command in privilege mode.



To save all Settings:

To save all the settings/configurations, type `copy running-config startup-config` command in privilege mode

```
router0#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

LAB TASKS

Task 1:

Choose a router in packet tracer. Name it with your registration # e.g., 21-CE-XXX. Set its time and date as today. Enable user mode and privileged mode passwords with your name.

Task 2:

Design a network having one router, a switch and two PC's as shown in the figure. Configure the router and assign the Class B IP addresses to the PC's and ping the default gateway. Also save the current configuration of the router.

Task 3:

Simulate the network which contains four PCs and two routers. Ensure communication between two routers.

Conclusion
